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In [1]:

```
import numpy as np
import pandas as pd

# 1. Create a DataFrame from a list
data = [
    [1, 'Smith', 50000],
    [2, 'Jones', 60000]
]

df = pd.DataFrame(data)
df.columns = ['Empd', 'Name', 'Salary']

print("Employee DataFrame:")
print(df)

# 2. Display DataFrame information
print("\nDataFrame Information:")
df.info()

# 3. Create an employee dataset
employee_data = {
    'emp id': [1, 2, 3, 4, 5, 6, 7],
    'name': [
        'SREE VARSSINI K S',
        'SREEMATHI B',
        'SREYA G',
        'SREYASKARI MULLAPUDI',
        'SRI AKASH U G',
        'SRI HARSHAVARDHANAN R',
        'SRI HARSHAVARDHANAN R'
    ],
    'salary': [5000, 6000, 7000, 5000, 8000, 3000, 6000]
}

df = pd.DataFrame(employee_data)

# 4. Display the first and last five rows
print("\nFirst Five Rows:")
print(df.head())

print("\nLast Five Rows:")
print(df.tail())

# 5. Display DataFrame information
print("\nEmployee DataFrame Information:")
df.info()

# 6. Select the salary column
print("\nSalary Column:")
print(df['salary'])

# 7. Calculate statistical measures
print("\nMean Salary:", df['salary'].mean())
print("Median Salary:", df['salary'].median())
```

```python
print("Mode Salary:", df['salary'].mode().tolist())
print("Salary Variance:", df['salary'].var())
print("Salary Standard Deviation:", df['salary'].std())

# 8. Display numerical summary
print("\nStatistical Summary:")
print(df.describe())

# 9. Display summary of all columns
print("\nComplete Summary:")
print(df.describe(include='all'))

# 10. Display column names
print("\nColumn Names:")
print(df.columns)

# 11. Convert DataFrame into a NumPy array
emp_array = df.values
print("\nEmployee Array:")
print(emp_array)

# 12. Create a DataFrame from the NumPy array
employee_DF = pd.DataFrame(emp_array, columns=df.columns)

print("\nFinal Employee DataFrame:")
print(employee_DF)
```

```
Employee DataFrame:
   Empd   Name  Salary
0     1  Smith   50000
1     2  Jones   60000

DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      str
 2   Salary  2 non-null      int64
dtypes: int64(2), str(1)
memory usage: 190.0 bytes

First Five Rows:
   emp id                  name  salary
0       1     SREE VARSSINI K S    5000
1       2           SREEMATHI B    6000
2       3               SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000

Last Five Rows:
   emp id                   name  salary
2       3                SREYA G    7000
3       4   SREYASKARI MULLAPUDI    5000
4       5          SRI AKASH U G    8000
5       6  SRI HARSHAVARDHANAN R    3000
6       7  SRI HARSHAVARDHANAN R    6000

Employee DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      str
 2   salary  7 non-null      int64
dtypes: int64(2), str(1)
memory usage: 410.0 bytes

Salary Column:
0    5000
1    6000
2    7000
3    5000
4    8000
5    3000
6    6000
Name: salary, dtype: int64

Mean Salary: 5714.285714285715
```

```
Median Salary: 6000.0
Mode Salary: [5000, 6000]
Salary Variance: 2571428.5714285714
Salary Standard Deviation: 1603.5674514745463

Statistical Summary:
         emp id       salary
count  7.000000     7.000000
mean   4.000000  5714.285714
std    2.160247  1603.567451
min    1.000000  3000.000000
25%    2.500000  5000.000000
50%    4.000000  6000.000000
75%    5.500000  6500.000000
max    7.000000  8000.000000

Complete Summary:
          emp id                    name       salary
count   7.000000                       7     7.000000
unique       NaN                       6          NaN
top          NaN  SRI HARSHAVARDHANAN R          NaN
freq         NaN                       2          NaN
mean    4.000000                     NaN  5714.285714
std     2.160247                     NaN  1603.567451
min     1.000000                     NaN  3000.000000
25%     2.500000                     NaN  5000.000000
50%     4.000000                     NaN  6000.000000
75%     5.500000                     NaN  6500.000000
max     7.000000                     NaN  8000.000000

Column Names:
Index(['emp id', 'name', 'salary'], dtype='str')

Employee Array:
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHANAN R' 3000]
 [7 'SRI HARSHAVARDHANAN R' 6000]]

Final Employee DataFrame:
  emp id                   name salary
0      1      SREE VARSSINI K S   5000
1      2            SREEMATHI B   6000
2      3                SREYA G   7000
3      4   SREYASKARI MULLAPUDI   5000
4      5          SRI AKASH U G   8000
5      6  SRI HARSHAVARDHANAN R   3000
6      7  SRI HARSHAVARDHANAN R   6000
```

In [ ]: